

Event records shall be cleared from the device for the device to recycle those entries for a future event. Each returned event record includes an event-log specific, nonzero, record handle that the host shall use when clearing events from the device's event log. The device shall maintain unique handles for every event that is placed in each event log.

In response to this command, the device shall return an overflow indicator when there are more events detected than could be stored in the specific event log. This indicator shall remain set until the host has consumed one or more events and called Clear Event Records to return the event handles to the device. When an event log overflows, the device shall retain all event records, in the specific event log, that occurred before the overflow event.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length
- Media Disabled
- Busy

Command Effects:

- None

Table 8-64. Get Event Records Input Payload

Byte Offset	Length in Bytes	Description
00h	1	Event Log: The specific device event log for which to retrieve the next event records. • 00h = Informational Event Log • 01h = Warning Event Log • 02h = Failure Event Log • 03h = Fatal Event Log • 04h = Dynamic Capacity Event Log • All other encodings are reserved

Table 8-65. Get Event Records Output Payload

Byte Offset	Length in Bytes	Description
00h	1	Flags • Bit[0]: Overflow: This bit shall be set by the device when errors occur that the device cannot log without overwriting an existing log event. When set, the Overflow Error Count, First Overflow Event Timestamp and Last Overflow Event Timestamp fields shall be valid. This indicator shall remain set until the host has consumed one or more events and returned the event handles to the device or cleared all the events using the Clear Event Records command. • Bit[1]: More Event Records: This bit shall be set by the device if there are more event records in this event log to retrieve than fit in the Get Event Records output payload. The host should continue to retrieve records using this command, until this indicator is no longer set by the device. • Bits[7:2]: Reserved.
1	1	Reserved
2	2	Overflow Error Count: The number of errors detected by the device that were not logged because of an overflow of this event log. The counter does not wrap if more errors occur than can be counted. A value of 0 indicates the device does not have a count of errors that occurred after the overflow was established. This field is only valid if the overflow indicator is set.
4	8	First Overflow Event Timestamp: The time of the first event that caused the overflow of the event log to occur. This field is only valid if the overflow indicator is set. The number of unsigned nanoseconds that have elapsed since midnight, 01-Jan-1970, UTC. If the device does not have a valid timestamp, return 0.
0Ch	8	Last Overflow Event Timestamp: The time of the last event that the device detected since the overflow of the event log occurred. This field is only valid if the overflow indicator is set. The number of unsigned nanoseconds that have elapsed since midnight, 01-Jan-1970, UTC. If the device does not have a valid timestamp, return 0.
14h	2	Event Record Count: The number of event records in the Event Records list. A value of 0 indicates that there are no more event records to return.
16h	0Ah	Reserved
20h	Varies	Event Records: A list of returned Event Records.

8.2.10.2.3 Clear Event Records (Opcode 0101h)

Clear Event Records provides a mechanism for the host to clear events that it has consumed from the device's Event Log.

If the host has more events to clear than space in the input payload, it shall use multiple calls to Clear Event Records to clear them all.

If the Event Log has overflowed, the host may clear all the device's stored event logs for the requested Event Log instead of explicitly clearing each event with the unique handle. Events shall be cleared in temporal order. The device shall verify the event record handles specified in the input payload are in temporal order. If the device detects an older event record that will not be cleared when Clear Event Records is executed, the device shall return the Invalid Handle return code and shall not clear any of the specified event records.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

- Retry Required
- Invalid Handle
- Invalid Payload Length
- Media Disabled
- Busy

Command Effects:

- Immediate Log Change

Table 8-66. Clear Event Records Input Payload

Byte Offset	Length in Bytes	Description
00h	1	Event Log: The specific device Event Log for which to clear the Event Records. • 00h = Informational Event Log • 01h = Warning Event Log • 02h = Failure Event Log • 03h = Fatal Event Log • 04h = Dynamic Capacity Event Log • All other encodings are reserved
01h	1	Clear Event Flags: If 0, the device shall clear the event records specified in the Event Record Handles list. • Bit[0]: Clear All Events: When set, the device shall clear all events that it currently has stored internally for the requested Event Log. Clear All Events is only allowed when the Event Log has overflowed; otherwise, the device shall return Invalid Input. • Bits[7:1]: Reserved.
02h	1	Number of Event Record Handles: The number of Event Record Handles in the Clear Event Records input payload. If Clear All Events is set, this shall be 0.
03h	3	Reserved
06h	Varies	Event Record Handles: A list of Event Record Handles the host has consumed and the device shall now remove from its internal Event Log store. These values are device specific and reported to the host in each event record using the Get Event Records command. All event record handles shall be nonzero value. A value of 0 shall be treated by the device as an invalid handle.

8.2.10.2.4 Get Event Interrupt Policy (Opcode 0102h)

Retrieve the settings for interrupts that are signaled for device events.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 8-67. Get Event Interrupt Policy Output Payload

Byte Offset	Length in Bytes	Description
00h	1	Informational Event Log Interrupt Settings: When enabled, the device shall signal an interrupt when the information event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved • Bits[7:4]: Interrupt Message Number: See definition below
01h	1	Warning Event Log Interrupt Settings: When enabled, the device shall signal an interrupt when the warning event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved • Bits[7:4]: Interrupt Message Number: See definition below
02h	1	Failure Event Log Interrupt Settings: When enabled, the device shall signal an interrupt when the failure event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved • Bits[7:4]: Interrupt Message Number: See definition below
03h	1	Fatal Event Log Interrupt Settings: When enabled, the device shall signal an interrupt when the fatal event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved • Bits[7:4]: Interrupt Message Number: See definition below
04h	1	Dynamic Capacity Event Log Interrupt Settings: When enabled, the device shall signal an interrupt when the Dynamic Capacity event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = Reserved — 11b = Reserved • Bits[3:2]: Reserved • Bits[7:4]: Interrupt Message Number: See definition below

Interrupt Message Number is defined as follows:

If Interrupt Mode = MSI/MSI-X:

- For MSI, interrupt message number indicates the offset between the base Message Data and the interrupt message that is generated. Hardware is required to update this field so that it is correct if the number of MSI Messages assigned to the Function changes when software writes to the Multiple Message Enable field in the Message Control register for MSI.

- For MSI-X, interrupt message number indicates which MSI-X Table entry is used to generate the interrupt message. The entry shall be one of the first 16 entries even if the Function implements more than 16 entries. The value shall be within the range configured by system software to the device. For a given MSI-X implementation, the entry shall remain constant.
- If both MSI and MSI-X are implemented, they are permitted to use different vectors, though software is permitted to enable only one mechanism at a time. If MSI-X is enabled, interrupt message number shall indicate the vector for MSI-X. If MSI is enabled or neither is enabled, interrupt message number shall indicate the vector for MSI. If software enables both MSI and MSI-X at the same time, interrupt message number is undefined.

If Interrupt Mode = FW Interrupt:

- Interrupt message number is the FW Interrupt vector the device uses to issue Firmware Notification via Event Firmware Notification (EFN) message.

8.2.10.2.5 Set Event Interrupt Policy (Opcode 0103h)

Change the settings for the interrupts that are signaled for device events. All event log interrupt settings shall be reset to 00b (No Interrupts) by the device on Conventional Reset.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- Immediate Policy Change

Table 8-68. Set Event Interrupt Policy Input Payload

Byte Offset	Length in Bytes	Description
00h	1	Informational Event Log Interrupt Settings: Specifies the settings for the interrupt when the information event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved. • Bits[7:4]: FW Interrupt Message Number: Specifies the FW interrupt vector the device shall use to issue the firmware notification. Valid only if Interrupt Mode = FW Interrupt.
01h	1	Warning Event Log Interrupt Settings: Specifies the settings for the interrupt when the warning event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved. • Bits[7:4]: FW Interrupt Message Number: Specifies the FW interrupt vector the device shall use to issue the firmware notification. Valid only if Interrupt Mode = FW Interrupt.
02h	1	Failure Event Log Interrupt Settings: Specifies the settings for the interrupt when the failure event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved. • Bits[7:4]: FW Interrupt Message Number: Specifies the FW interrupt vector the device shall use to issue the firmware notification. Valid only if Interrupt Mode = FW Interrupt.
03h	1	Fatal Event Log Interrupt Settings: Specifies the settings for the interrupt when the fatal event log transitions from having no entries to having one or more entries. • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = FW Interrupt (EFN VDM) — 11b = Reserved • Bits[3:2]: Reserved. • Bits[7:4]: FW Interrupt Message Number: Specifies the FW interrupt vector the device shall use to issue the firmware notification. Valid only if Interrupt Mode = FW Interrupt.
04h	1	Dynamic Capacity Event Log Interrupt Settings: Specifies the settings for the interrupt when the Dynamic Capacity Event Log transitions from having no entries to having one or more entries.¹ • Bits[1:0]: Interrupt Mode: — 00b = No interrupts — 01b = MSI/MSI-X — 10b = Reserved — 11b = Reserved • Bits[7:2]: Reserved

1. This field is not included in the Minimum Input Payload Length.

8.2.10.2.6 Get MCTP Event Interrupt Policy (Opcode 0104h)

Retrieve the settings for interrupts that are signaled for component events over MCTP management media. When notifications are enabled for a particular log on a component, the component may issue no more than 1 Event Notification for that log type when the log contents transition from zero entries to one or more entries. The FM must be able to receive at least 1 Event Notification message from the component for each log for which interrupts have been enabled.

When notifications are enabled for Background Operation completion, the component shall issue an Event Notification upon the completion of every Background Operation. This setting shall be ignored if the component does not support Background Operations.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 8-69. Payload for Get MCTP Event Interrupt Policy Output, Set MCTP Event Interrupt Policy Input, and Set MCTP Event Interrupt Policy Output

Byte Offset	Length in Bytes	Description
00h	2	Event Interrupt Settings: Bitmask indicating whether event notifications are enabled (1) or disabled (0) for a particular event. • Bit[0]: New uncleared Informational Event Log record(s) • Bit[1]: New uncleared Warning Event Log record(s) • Bit[2]: New uncleared Failure Event Log record(s) • Bit[3]: New uncleared Fatal Event Log record(s) • Bit[4]: New uncleared Dynamic Capacity Event Log record(s) • Bits[14:5]: Reserved • Bit[15]: Background Operation completed

8.2.10.2.7 Set MCTP Event Interrupt Policy (Opcode 0105h)

Change the settings for the interrupts that are signaled for component events. The receiver may capture the address of the requesting component in a transport-specific way. The subsequent enabled events shall be sent to that address.

Interrupts shall only be generated for events that occur after this command is received by the component with input parameters that enable logging for the corresponding log type. Components should immediately terminate the retransmission of Event Notification requests that have not been acknowledged if a request of this type has been received with input parameters that disable logging for the corresponding log type.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required

- Invalid Payload Length

Command Effects:

- Immediate Policy Change

Input and output payloads for this command are documented in [Table 8-69](#).

8.2.10.2.8 Event Notification (Opcode 0106h)

This command is used by a CXL component to send notifications from its MCTP-based CCIs to the agent that issued the Set MCTP Event Interrupt Policy request (see [Section 8.2.10.2.7](#)). It is only sent by CXL components. Any commands of this type received by a CXL component's MCTP-based CCI shall be silently discarded.

This command is a notification that there are new events to be read from the event logs specified in the "Event" payload field. A notification for log-based events is triggered when both of the following conditions are met: the log transitions from having no entries to having one or more entries and MCTP interrupts have been enabled with the Set MCTP Event Interrupt Policy command. Log entries are cleared with the Clear Event Logs command. After clearing log entries, the Get Event Records command should be used to confirm that all entries have been cleared from the log to ensure that the notification trigger has been re-armed. A notification for Background Operations is sent when a Background Operation completes.

The recipient of this message may optionally acknowledge a notification by returning a response.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 8-70. Event Notification Input Payload

Byte Offset	Length in Bytes	Description
00h	2	Event: Bitmask that indicates the event. • Bit[0]: Informational Event Log has uncleared record(s) • Bit[1]: Warning Event Log has uncleared record(s) • Bit[2]: Failure Event Log has uncleared record(s) • Bit[3]: Fatal Event Log has uncleared record(s) • Bit[4]: Dynamic Capacity Event Log has uncleared record(s) • Bits[14:5]: Reserved • Bit[15]: Background Operation completed

8.2.10.2.9 GFD Enhanced Event Notification (Opcode 0107h)

This VDM payload format is used by a GFD component to send notifications to the FM. It is only sent by GFD components.

This is a notification to the FM that there are new events to be read from the event logs specified in the "Event" payload field. A notification for log-based events is triggered when both of the following conditions are true:

- Log transitions from having no entries to having one or more entries
- MCTP interrupts have been enabled with the Set MCTP Event Interrupt Policy command

Log entries are cleared with the Clear Event Logs command. After clearing log entries, the FM should use the Get Event Records command to confirm that all entries have been cleared from the log to ensure that the notification trigger has been re-armed. A notification for Background Operations is sent when a Background Operation completes.

The FM acknowledges a notification by returning a response.

GFDs may use the general Event Notification (see [Section 8.2.10.2.8](#)) mechanism to signal the FM, but are highly encouraged to use the Enhanced Event Notification format to deliver more-timely details about the specific event that triggered this notification.

Because GFDs do not have Event Logs per host, this enhanced event payload format shall be used by GFDs to signal individual host GAEs of specific GFD events that require host notification, such as Dynamic Capacity Change Events or asynchronous write failures.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 8-71. Enhanced Event Notification Input Payload (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	2	PBR_ID: PBR_ID of the GFD that is generating this Notification.
02h	2	Event: Bitmask that indicates the event logs that need attention. • Bit[0]: Informational Event Log has uncleared record(s) • Bit[1]: Warning Event Log has uncleared record(s) • Bit[2]: Failure Event Log has uncleared record(s) • Bit[3]: Fatal Event Log has uncleared record(s) • Bit[4]: Dynamic Capacity Event Log has uncleared record(s)¹ • Bits[14:5]: Reserved • Bit[15]: Background Operation completed
04h	1	GFD Notification Type: Encoded as per Table 8-72 .
05h	1	Notification Type Trigger: Details of root cause of error or event that triggered this notification. The encoding of this field is dependent on the GFD Notification Type. See Table 8-73 for these encodings.
06h	2	RPID: Requester PID of request that was being processed for which an error was detected and/or this event notification triggered. If no specific request can be identified, the RPID shall not be marked valid. • Bits[11:0]: RPID of Request • Bits[15:12]: Reserved
08h	8	DPA: Device Physical Address decoded from requester's provided HPA. See Section 7.7.2.4 . DPA value is valid only if the Request Associated bit is set AND the DPA valid bit is set.

Table 8-71. Enhanced Event Notification Input Payload (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
10h	8	HPA: Host Physical Address extracted from request. See Section 7.7.2.4 . HPA value is valid only if the Request Associated bit is set AND the HPA valid bit is set.
18h	2	Memory Group ID (GrpID): The GrpID value decoded from the DPA. GFDs allocate the device's DPA space to Memory Groups (see Section 7.7.2.5), which have a handle (GrpID). Every DPA that has memory allocated to back it must resolve to a defined GrpID. GrpID value is valid only if the Request Associated bit is set AND the Memory Group valid bit is set. Note: If this Enhanced Event Notification is sent to a non-FM PBR_ID, the Memory Group valid bit shall be false and this Memory Group ID field is reserved.
1Ah	1	Memory Req Opcode: The opcode extracted from the memory request. - Bits[3:0]: Memory Req Opcode (see Table 3-35) - Bits[7:4]: Reserved Memory Req Opcode value is valid only if the Request Associated bit is set AND the Memory Req Opcode bit is set.
1Bh	1	Reserved
1Ch	2	Flags: Various valid and condition flags. - Bit[0]: Reserved - Bit[1]: Request Associated: If 1, device has associated a single memory request with the triggering event, and the request's details may be valid, per remaining flags - Bit[2]: DPA Valid: If 1, DPA field is valid - Bit[3]: HPA Valid: If 1, HPA field is valid - Bit[4]: Memory Group ID Valid: If 1, GrpID value is valid - Bit[5]: Memory Req Opcode Valid: If 1, Memory Req Opcode value is valid - Bits[15:6]: Reserved
1Eh	2	Event Record Handle: The event log unique handle for the associated event record.
20h	8	Timestamp: Event Record Timestamp per Table 8-55 .

Table 8-72. GFD Notification Type Values (Sheet 1 of 2)

Notification Type	Description	Encoding
No-Op	Reserved for testing of Enhanced Event Notification delivery process. The FM shall ignore this notification.	00h
Standard Event Notification	Standard Event Notification. This command is a notification to the FM that there are new events to be read from the event logs specified in the Event payload field. No more specific Notification Type is applicable; however, some of the Enhanced Event Notification Payload fields may be valid.	01h
GFD Access Error	Memory request failed. Device could not validate Host SPID has access permissions to the specific HPA in the request, or could not decode a valid DPA from the supplied HPA.	02h
GFD Unrecognized Request	Device does not recognize the request Opcode.	03h
GFD Unexpected Packet	Device received a Response packet, a notification packet, a request type reserved for FM, etc., for which it has no associated context.	04h
GFD Invalid Input	Request fields contained invalid values or incompatible settings.	05h
GFD Persistent Flush Failure	Device's attempts to flush Persistent Memory Write Buffers have failed. This notification may be associated with the state of the Dirty Shutdown Count. See Table 8-148 .	06h

Table 8-72. GFD Notification Type Values (Sheet 2 of 2)

Notification Type	Description	Encoding
GFD Execution Failure: Non-Fatal	Device failed to successfully execute a Memory Request, device state is Non-Fatal and device is capable of processing subsequent Memory Requests.	07h
GFD Execution Failure: Fatal	Device failed to successfully execute a Memory Request, device state is Fatal and device is not capable of processing subsequent Memory Requests.	08h
GFD Health Alerts	Device Health Events have been logged and one or more notification thresholds have been reached.	09h
GFD Extent List Change	Dynamic Capacity associated with this host may have changed. Host must reread the Extent List of this GFD.	0Ah
Reserved	Reserved	0Bh - FFh

Table 8-73. GFD Notification Type Triggers (Sheet 1 of 2)

Notification Type	Notification Type Trigger Values
No-Op	Bits[7:0]: Reserved
Standard Event Notification	Bits[7:0]: Reserved
GFD Access Error	- Bit[0]: HPA-to-DPA mapping is non-existent, HPA was not claimed by any of the host's decoders - Bit[1]: DPA-to-HPA snoop mapping failed, Back-Invalidate HPA unresolved - Bit[2]: DPA to Memory Group ID (GrpID) failed, GrpID is undefined - Bit[3]: SPID access permission denied, GrpAccVec[GrpID] is not set in SAT[SPID] - Bits[7:4]: Reserved
GFD Unrecognized Request	- Bit[0]: Not a valid memory request opcode - Bit[1]: Not a supported request type - Bits[7:2]: Reserved
GFD Unexpected Packet	Bits[7:0]: Reserved
GFD Invalid Input	Bits[7:0]: Reserved
GFD Persistent Flush Failure	- Bit[0]: Dirty Shutdown Counter has incremented (detected after reset) - Bit[1]: Internal error detected during the flush (detected before reset) - Bit[2]: Media error detected during flush (detected before reset) - Bits[7:3]: Reserved
GFD Execution Failure: Non-Fatal	- Bit[0]: DPA-to-HPA snoop mapping failed, Back-Invalidate HPA unresolved. Error occurred using a sharer's PBR_ID, not the requester's. - Bit[1]: Back-Invalidate request received unsuccessful response - Bit[2]: Back-Invalidate request received no response - Bit[3]: Transient internal uncorrectable ECC error detected - Bits[7:4]: Reserved

Table 8-73. GFD Notification Type Triggers (Sheet 2 of 2)

Notification Type	Notification Type Trigger Values
GFD Execution Failure: Fatal	- Bit[0]: Persistent internal uncorrectable ECC error detected - Bit[1]: Persistent media access error - Bits[7:2]: Reserved
GFD Health Alerts	Health Alerts: Indicators of which alert thresholds have been exceeded. - Bit[0]: When set, the Life Used Programmable Warning Threshold has tripped - Bit[1]: When set, the Device Over-Temperature Programmable Warning Threshold has tripped - Bit[2]: When set, the Device Under-Temperature Programmable Warning Threshold has tripped - Bit[3]: When set, the Corrected Volatile Memory Error Programmable Warning Threshold has tripped - Bit[4]: When set, the Corrected Persistent Memory Error Programmable Warning Threshold has tripped - Bits[7:5]: Reserved

8.2.10.2.10 GFD to GAE Enhanced Event Notification (Opcode 0108h)

This command is used by a GFD component to send asynchronous event notifications to a host GAE. It is sent only by GFD components.

Because a GFD does not keep logs on a per-host basis, a standard Event Notification (see [Section 8.2.10.2.8](#)) will not suffice when signaling a host of an asynchronous error or event that has occurred as a result of processing a host's request. The host does not have access to the global event logs of the GFD.

This enhanced event payload format shall be used by GFDs to signal individual host GAEs of specific GFD events that require host attention, mainly to report asynchronous write failures. The enhanced event notification payload contains relevant fields from the host's failed memory request.

This Event Notification is transported to the target host GAE using the GAM VDM, as described in [Section 3.1.11.7](#). Its payload is logged in the GAM Buffer, a circular buffer in host memory

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

Command Effects:

- None

Table 8-74. Enhanced Event Notification Input Payload (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	2	PBR_ID: PBR_ID of the GFD that is generating this Notification.
02h	1	GFD Notification Type: Encoded as per Table 8-72 .
03h	1	Memory Req Opcode: The Opcode extracted from the memory request. - Bits[3:0]: Memory Req Opcode (see Table 3-35) - Bits[7:4]: Reserved Memory Req Opcode value is valid only if the Memory Req Opcode bit is set.

Table 8-74. Enhanced Event Notification Input Payload (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
04h	2	Flags: Various valid and condition flags. • Bit[0]: Reserved • Bit[1]: Hardwire to 1 • Bit[2]: DPA Valid: If 1, DPA field is valid • Bit[3]: HPA Valid: If 1, HPA field is valid • Bit[4]: Reserved • Bit[5]: Memory Req Opcode Valid: If 1, Memory Req Opcode value is valid • Bits[15:6]: Reserved
06h	2	RPID: Requester PID of request that was being processed for which an error was detected and/or this event notification triggered. This PID must match one in use within this host's domain. • Bits[11:0]: RPID of Request • Bits[15:12]: Reserved
08h	8	DPA: Device Physical Address decoded from requester's provided HPA. See Section 7.7.2.4 . DPA value is valid only if the DPA valid bit is set.
10h	8	HPA: Host Physical Address extracted from request. See Section 7.7.2.4 . HPA value is valid only if the HPA valid bit is set.
18h	8	Timestamp: Event Record Timestamp per Table 8-55 , Common Event Record Format.

8.2.10.2.11 Get GAM Buffer (Opcode 0109h)

This command is used by a host to read the local pointer used for the GAM Buffer, as described in [Section 3.1.11.7](#) and [Section 8.2.10.2.10](#). It is supported only by GAEs.

The format of the GAM Buffer entries is defined in [Table 8-73](#).

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

Command Effects:

- None

Table 8-75. Get GAM Buffer Response Payload

Byte Offset	Length in Bytes	Description
00h	8	• Bit[0]: Valid: Indicates whether the GAM Buffer Address is valid (1) or invalid (0) • Bit[1]: Buffer Overflow: When set, indicates that the GAE was unable to log a received GAM payload to the GAM Buffer because there were no available entries • Bits[4:2]: Reserved • Bits[51:5]: GAM Buffer Address[51:5]: Address to which the GAM payload will be written by the GAE • Bits[55:52]: Reserved • Bits[63:56]: Head Index: Indicates the index at which the GAE will log the next received GAM payload

8.2.10.2.12 Set GAM Buffer (Opcode 010Ah)

This command is used by a host to update the local pointer used for the GAM Buffer, as described in [Section 3.1.11.7](#) and [Section 8.2.10.2.10](#). It is supported only by GAEs.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

Command Effects:

- None

Table 8-76. Set GAM Buffer Address Request Payload

Byte Offset	Length in Bytes	Description
00h	8	• Bit[0]: Valid: Indicates whether the GAM Buffer Address is valid (1) or invalid (0) • Bit[1]: Clear Overflow: When set, clears the "Buffer Overflow" bit reported in Get GAM Buffer • Bits[4:2]: Reserved • Bits[51:5]: GAM Buffer Address[51:5]: Address to which the GAM payload will be written by the GAE • Bits[55:52]: Reserved • Bits[63:56]: Tail Index: Indicates the index up to which the host has read GAM Buffer entries

8.2.10.3 Firmware Update

FW Update is an optional feature for devices to provide a mechanism to update the FW. If supported, the Get FW Info command and the Transfer FW command shall be supported; the Activate FW command is optional.

FW refers to an FW package that may contain multiple FW images. The management of multiple FW images or the FW on multiple controllers is the responsibility of the device and beyond the scope of this specification. Product vendors can implement any means of managing multiple FW images, provided those means do not conflict with or alter the specifications described herein.

The number of FW slots the device supports is vendor specific, up to four. The minimum FW slots supported shall be two, one slot for the active FW and one slot for an FW package that is staged for activation. Only one slot may be active at a time and only one slot may be staged for activation at a time. The staged FW will transition to the active FW on next reset (as specified in the Transfer FW command effects) or upon successful execution of the Activate FW command, if supported. Other FW packages that are fully transferred and stored in a slot shall persist across power cycles.

8.2.10.3.1 Get FW Info (Opcode 0200h)

Retrieve information about the device FW.

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required

- Invalid Payload Length

Command Effects:

- None

Table 8-77. Get FW Info Output Payload

Byte Offset	Length in Bytes	Description
00h	1	FW Slots Supported: The number of FW slots the device supports, up to four. The minimum FW slots supported shall be two, one slot for the active FW and one slot for an FW that is staged for activation.
01h	1	FW Slot Info: Indicates the active and staged FW slots. • Bits[2:0]: The slot number of the active FW version. Only one slot may be active at a time. 0 is an illegal value. Values greater than "FW Slots Supported" are also illegal. • Bits[5:3]: The slot number of the FW that will be activated on the next reset or on successful execution of the Activate FW command, if supported. If 0, no FW is currently staged for activation. Refer to the FW Activation Capabilities to determine whether the FW can be activated at runtime. If the FW fails to activate, the device shall fall back to the previously active FW. Only one slot may be staged for activation at a time. Values greater than "FW Slots Supported" are illegal. • Bits[7:6]: Reserved
02h	1	FW Activation Capabilities: Defines the capabilities supported by the device for activating a new FW without a reset. • Bit[0]: When set, the currently active FW on the device supports online FW activation with the Activate FW command • Bit[1]: When set, per slot FW activation information is available. When this bit is 0, then Bit[5:2] at this offset are Reserved. • Bit[2]: When set, FW package in Slot 1 is online activatable. If 0, either no FW is available in Slot 1, or the FW package in Slot 1 is not online activatable. • Bit[3]: When set, FW package in Slot 2 is online activatable. If 0, either no FW is available in Slot 2, or the FW package in Slot 2 is not online activatable. • Bit[4]: When set, FW package in Slot 3 is online activatable. If 0, either no FW is available in Slot 3, or the FW package in Slot 3 is not online activatable. • Bit[5]: When set, FW package in Slot 4 is online activatable. If 0, either no FW is available in Slot 4, or the FW package in Slot 4 is not online activatable. • Bits[7:6]: Reserved
03h	13	Reserved
10h	16	Slot 1 FW Revision: Contains the revision of the FW package in Slot 1 formatted as an ASCII string. If there is no FW in Slot 1, this field shall be cleared to 0.
20h	16	Slot 2 FW Revision: Contains the revision of the FW package in Slot 2 formatted as an ASCII string. If there is no FW in Slot 2, this field shall be cleared to 0.
30h	16	Slot 3 FW Revision: Contains the revision of the FW package in Slot 3 formatted as an ASCII string. If there is no FW in Slot 3 or the device does not support 3 or more FW slots, this field shall be cleared to 0.
40h	16	Slot 4 FW Revision: Contains the revision of the FW package in Slot 4 formatted as an ASCII string. If there is no FW in Slot 4 or the device does not support 4 FW slots, this field shall be cleared to 0.

8.2.10.3.2 Transfer FW (Opcode 0201h)

Transfer all or part of an FW package from the caller to the device. FW packages and all parts of each FW package shall be 128-byte aligned.

If the FW package is transferred in its entirety, the caller makes one call to Update FW with Action = Full FW Transfer.

If an FW package is transferred in parts, the caller makes one call to Transfer FW with Action = Initiate FW Transfer, zero or more calls with Action = Continue FW Transfer, and one call with Action = End Transfer or Abort Transfer. The FW package parts shall be transferred in order; otherwise, the device shall return the FW Transfer Out of Order return code. Back-to-back retransmission of any FW package part is permitted during a transfer.

IMPLEMENTATION NOTE

Example of FW image transmissions:

The following example sequence transfers 380h bytes worth of an FW image. Note that the Offset field is expressed in multiples of 80h Bytes:

1. Transfer FW (Action=Initiate FW Transfer, Slot=x, Offset=0, 100h B package).
2. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=2, 100h B package).
3. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=4, 100h B package).
4. Transfer FW (Action=End Transfer, Slot=1, Offset=6, 80h B package).

The following example is also a valid in-order sequence. It also transfers 380h bytes worth of an FW image. Note that Bytes 100h-1FFh are transmitted twice, but are sent back-to-back and are therefore legal:

1. Transfer FW (Action=Initiate FW Transfer, Slot=x, Offset=0, 100h B package).
2. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=2, 100h B package).
3. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=2, 100h B package).
4. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=4, 100h B package).
5. Transfer FW (Action=End Transfer, Slot=1, Offset=6, 80h B package).

The following example is an invalid in-order sequence:

1. Transfer FW (Action=Initiate FW Transfer, Slot=x, Offset=0, 100h B package).
2. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=2, 100h B package).
3. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=3, 180h B package).
4. Transfer FW (Action=End Transfer, Slot=1, Offset=6, 80h B package).

The following example is also an invalid in-order sequence:

1. Transfer FW (Action=Initiate FW Transfer, Slot=x, Offset=0, 100h B package).
2. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=2, 100h B package).
3. Transfer FW (Action=Continue FW Transfer, Slot=x, Offset=5, 80h B package).
4. Transfer FW (Action=End Transfer, Slot=1, Offset=6, 80h B package).

If an FW package transfer that initiated on a Mailbox CCI is interrupted by a Conventional or CXL reset, the FW package transfer shall be aborted by the device. If a FW package transfer is aborted prior to the entire FW package being successfully stored on the device, the device shall require the FW package transfer to be started from the beginning of the FW package.

Only one FW package may be transferred to a single CCI at a time. The device shall return the FW Transfer in Progress return code if it receives a Transfer FW command with Action = Full FW Transfer or Action = Initiate FW Transfer until the current FW package transfer is completed or aborted.

In case of a device with multiple CCIs, the device shall return the FW Transfer in Progress return code if it receives a Transfer FW request on any other CCI than the CCI that initiated the current FW package transfer until that transfer is completed or aborted. The timeout between FW package parts is 30 seconds, after which point the device shall consider the current FW package transfer aborted.

Once the entire FW package is fully transferred to the device, the device shall verify the FW package and store it in the specified slot. Verification of the FW package is vendor specific.

In support of confidential computing, if the device has been locked while utilizing secure CXL TSP interfaces and the device was not enabled for FW update after being locked, the device shall reject the Transfer FW command by returning Invalid Security State status. See [Section 11.5](#) for details on locking a device and locked device behavior.

Possible Command Return Codes:

- Success
- Background Command Started
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length
- Media Disabled
- Busy
- FW Transfer in Progress
- FW Transfer Out of Order
- FW Verification Failed
- Invalid Slot
- Aborted
- Invalid Security State

Command Effects:

- Configuration Change after Cold Reset
- CEL[11:10] Valid
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Background Operation
- Request Abort Background Operation Command Supported

Table 8-78. Transfer FW Input Payload

Byte Offset	Length in Bytes	Description
00h	1	Action: Specifies the stage of the FW package transfer. 00h = Full FW Transfer 01h = Initiate FW Transfer 02h = Continue FW Transfer 03h = End Transfer 04h = Abort Transfer - All other encodings are reserved
01h	1	Slot: Specifies the FW slot number to store the FW once the transfer is complete and the FW package is validated. Shall not be the active FW slot; otherwise, the Invalid Slot return code shall be returned. Valid only if Action = Full transfer or End Transfer; otherwise, ignored.
02h	2	Reserved
04h	4	Offset: The byte offset in the FW package data. Expressed in multiples of 128 bytes. Ignored if Action = Full FW Transfer.
08h	78h	Reserved
80h	Varies	Data: The FW package data.

8.2.10.3.3 Activate FW (Opcode 0202h)

Activate FW is an optional command to make a FW previously stored on the device with the Transfer FW command, the active FW. Devices may support activating a new FW while online or on reset, as indicated by the Get FW Info command.

If a device supports online firmware activation, this command may be executed as a background command as indicated by the command return code.

If the new FW fails to online activate, the device shall roll back to the previous FW, if possible. A cold reset may be required to restore the operating state of the FW on activation failure.

If Software attempts to online activate a FW package that is not online activatable, the device shall return an error code. It is strongly recommended that the device return Activation Failed, FW Rolled Back in this scenario.

Online FW activation does not affect the data, the metadata bits (if present) and Extended Metadata (if present), that are stored by the device. In addition, CXL requests are correctly processed during online firmware activation.

In support of confidential computing, if the device has been locked while utilizing secure CXL TSP interfaces and the device was not enabled for FW update after being locked, the device shall reject this command by returning Invalid Security State status. See [Section 11.5](#) for details on locking a device and locked device behavior.

Possible Command Return Codes:

- Success
- Background Command Started
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

- Busy
- Activation Failed, FW Rolled Back
- Activation Failed, Cold Reset Required
- Invalid Slot
- Aborted
- Invalid Security State

Command Effects:

- Configuration Change after Cold Reset
- CEL[11:10] Valid
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Background Operation
- Request Abort Background Operation Command Supported

Table 8-79. Activate FW Input Payload

Byte Offset	Length in Bytes	Description
00h	1	Action: Specifies the activation method. • 00h = Online • 01h = On the next reset (as specified in the command effects) • All other encodings are reserved
01h	1	Slot: Specifies the FW slot number to activate. Shall not be the active FW slot; otherwise, the Invalid Slot return code shall be returned.

8.2.10.4 Timestamp

Timestamp is an optional setting that enables the host to set a time value in the device to correlate the device timer with the system time. The use of the timestamp is beyond the scope of this specification. The accuracy of the timestamp after it is set may be affected by vendor specific factors. Therefore, the timestamp shouldn't be used for time sensitive applications. Although the format of the timestamp is in nanoseconds, the resolution of time maintained by the device is implementation specific, so software shall not assume a device provides nanosecond resolution.

8.2.10.4.1 Get Timestamp (Opcode 0300h)

Get the timestamp from the device. Timestamp is initialized via the Set Timestamp command. If the timestamp has never been set, the output shall be 0.

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 8-80. Get Timestamp Output Payload

Byte Offset	Length in Bytes	Description
00h	8	Timestamp: The current device timestamp which represents the value set with the Set Timestamp command plus the number of nanoseconds that have elapsed since the timestamp was set.

8.2.10.4.2 Set Timestamp (Opcode 0301h)

Set the timestamp on the device. It is recommended that the host set the timestamp after every Conventional or CXL Reset. Otherwise, the timestamp may be inaccurate.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- Immediate Policy Change

Table 8-81. Set Timestamp Input Payload

Byte Offset	Length in Bytes	Description
00h	8	Timestamp: The number of unsigned nanoseconds that have elapsed since midnight, 01-Jan-1970, UTC.

8.2.10.5 Logs

Commands to return device-specific logs.

8.2.10.5.1 Get Supported Logs (Opcode 0400h)

Retrieve the list of device-specific logs (identified by UUID) and the maximum size of each Log.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None